RESEARCH ARTICLE CLASS PROJECT

Lifespan is Effected by Military Survice

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Abstract

Purpose: This project describes the lifespan in cemetery records and examines the variation by military service.

Methods: Recorded available birth and death information from gravestones and classified the military status of males using the visible evidence on the gravestones. Each name was recorded along with the date of thier birth, the date of their death, their gender, and if they served in the military.

Results: I have concluded that, on average, males enlisted in the military generally had longer lifespans than males who were not enlisted in the military.

Conclusions: The observed pattern of this sample suggests that military service increases lifespan on average, but this could potentially be limited due to the fact that most of the servicemen buried here simply were the ones who survived the wars they fought in and lived normal lives afterwards.

Keywords: cemetery records; lifespan; survivorship; historical demography; military status

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Background

Cemetery records provide individual birth and death dates that can be used to describe lifespan among people represented in a burial ground. These records are a non-random sample: cemetery access, preservation of markers, migration, family burial practices, and the choice of graves to record may all affect which people appear in the dataset. The analyses below therefore describe the collected sample and should not be treated as direct estimates of lifespan for an entire town or historical population.

Hypotheses

Military service and lifespan: I hypothesize that male records marked as military service will show lower lifespan than records not marked for service because of drafting, exposure to chemical agents, and the dangers associated with military lifestyle.

Methods

Field site and sampling

- **Where did you go?** We went to the Sunset Hills Cemetary in Valdosta, Georgia located next to the VSU campus. I stayed around the eastern end of the cemetary and wrote down the information on the headstones.
- **What date and time of day did you collect records?** We went on Thursday October 8th, 2026, from around 2:15 PM to around 3:00 PM.
- **What part of the cemetery did you sample?** I entered around the southestern end and stayed around the eastern to northeastern part of the cemetary.
- **How did you choose graves?** I chose gravestones at random initially, but then went along down the rows, only skipping over the gravestones that were illegible or had no data on them.
- **How did you handle nearby graves and family plots?** I wrote down names next to each other if they were part of the same family, keeping them together before moving onto another last name.

Recorded information and variable definitions

- **What information did you record from each marker?** I recorded names, birth date, death date, gender, and military status.
- **How was lifespan calculated?** I calculated the lifespan by subtracting the death year by the birth year and then determined if they passed before or after their birthday within the year.
- **How did you determine marital status?** I did not record marital status in my data due to being unable to determine from the data the gravestones provided.
- **How did you determine military service?** The gravestones stated whether or not the deceased individual was a veteran along with their rank. Those that didn't state the individual was a veteran, I assumed they were not in the military.
- **How did you determine sex or gender?** I determined the gender of the individual based on their names and if they had a masculine or feminine name.
- **How were missing, illegible, or ambiguous records handled?** I skipped over the greavestones with illegible or missing data and did not record them.

Dataset and analysis

- **How many records were collected by you?** n = 25
- **How many records were included in the final analysis?** n = 702
- **Were records aggregated with the class dataset?** Yes. I noticed some duplicates involved in the data because there were many students involved in collecting the data, so some graves were accidentally repeated.
- **What summaries or plots were produced?** The plot which was produced showed a fairly similar result between males that had joined the military and males that did not join the military, the left part of the plot showing military males had higher survival chances which began to fluctuate as the ages increased.
- **What are the limits of the sample?** One of the main limitations of the sample is the minimum age of military service being 18, meaning that there are no military males which passed before their 18th birthday, increasing the mean age of military males compared to non-military males. Another limitation is that most of the servicemen buried in this cemetary survived the war and lived normal lives, whereas most of the serviemen who were killed on duty were buried in the Arlington National Cemetary.

Results

Replace each image placeholder with a PNG of the matching plot from the final class dataset. Keep figures in the `survivorship_curves` and `cemetery_lifespan_report_files` folder, or update the image path in the Markdown. The figure captions are editable and should describe the plotted groups, axes, and relevant reference lines.

Summarize the data both by referencing Figure 1 and by writing out what the table is displaying specific to your particular research hypothesis. For instance, if you are interested only in marriage status and sex, then report the mean and standard deviation of males and females that are married vs unmarried; you do not need to discuss military status unless this is part of your research question/hypothesis.

The summary table in Figure 1 shows substantial differences in recorded lifespan by sex in this sample. Males had a mean lifespan of 59.0 years (SD = 20.4; n = 3), compared with 29.7 years (SD = 7.1; n = 3) for females. The marital-status breakdown is more tentative because each subgroup contains only one or two records. Among males, the two married individuals had a mean lifespan of 70.5 years (SD = 6.4), while the one unmarried individual lived 36 years. Among females, the one married individual lived 36 years, compared with a mean of 26.5 years (SD = 6.4) for the two unmarried individuals. These small subgroup sizes make the values descriptive; they do not support broad conclusions about sex or marital status.

The sex-specific survivorship curves in Figure 2 show a

Lifespan by sex

Sex	n	Mean	SD	Median	Minimum	Maximum
f	342	71.1	21.9	76	0	105
m	410	62.2	23.0	67	0	114

Lifespan by sex and marital status

Sex	Married	n	Mean	SD	Median	Minimum	Maximum
f	n	164	69.0	25.9	76	0	105
f	y	178	73.2	17.2	77	16	101
m	n	240	57.8	25.8	64	0	97
m	y	170	68.5	16.4	71	25	114

Figure 1: Summary of lifespan by sex and by marrital status for each sex.

different pattern: the plotted means are 61.1 years for males and 57.3 years for females. The male curve remains higher through much of the earlier and middle lifespan, while the female curve reaches the 50% survival point later. The curves are close and change relative position, so the figure suggests that the sex difference depends on which part of the lifespan distribution is considered. The median markers indicate where each group's observed curve reaches 50% surviving; they are distinct from the mean values shown in the legend.

The within-sex comparisons in Figure 3 show higher plotted mean lifespans for married than unmarried individuals in both panels. For males, the plotted means are 64.3 years for married individuals and 55.4 years for unmarried individuals. For females, they are 64.3 and 46.3 years, respectively. The curves also place the 50% survival point later for the married groups. These are descriptive differences and should not be interpreted as evidence that marriage caused longer lifespan; other factors and the sizes and composition of the groups could contribute.

Summary of observed results: [Describe the main pattern, sample size for each comparison, and any uncertainty or missing-data issue. Refer to the figures by number. Do not infer causation from group differences.]

Discussion

Write a paragraph or two, citing your sources to prove the results of your data. **Sex and lifespan.** Research on sex differences in longevity points to a combination of biological, behavioral, and social influences rather than a single explanation. Austad and Fischer review evidence on sex differences in lifespan and aging across species, including humans (Austad & Fischer, 2016). In this cemetery sample, the observed female and male curves [describe the pattern] and [support / do not support] the prediction that [state your hypothesis]. The result should be interpreted in the context of this cemetery's birth cohorts and sampling process; it is not by itself an estimate of a general sex difference in lifespan.

Marital status and social relationships. Marriage can correlate with social support, health behaviors, household resources, and social integration, but people's health can also influence whether they marry or remain married. A large meta-analysis found associations between unmarried status and mortality, with some associations differing by sex (Wang *et al.*, 2020); a broader meta-analysis linked social relationships with mortality risk (Holt-Lunstad *et al.*, 2010). In this dataset, the comparison of married and not-married records among [men / women / both] showed [describe results]. These cemetery records do not measure relationship quality, social contact, or the timing and duration of marriage, so marital status should be treated as an imperfect indicator rather than a direct measure of socialization or a cause of longevity.

Historical changes in lifespan and childhood survival. Life expectancy at birth is strongly affected by deaths in infancy and childhood, so a low historical average does not mean that every adult died young. Improvements in clean

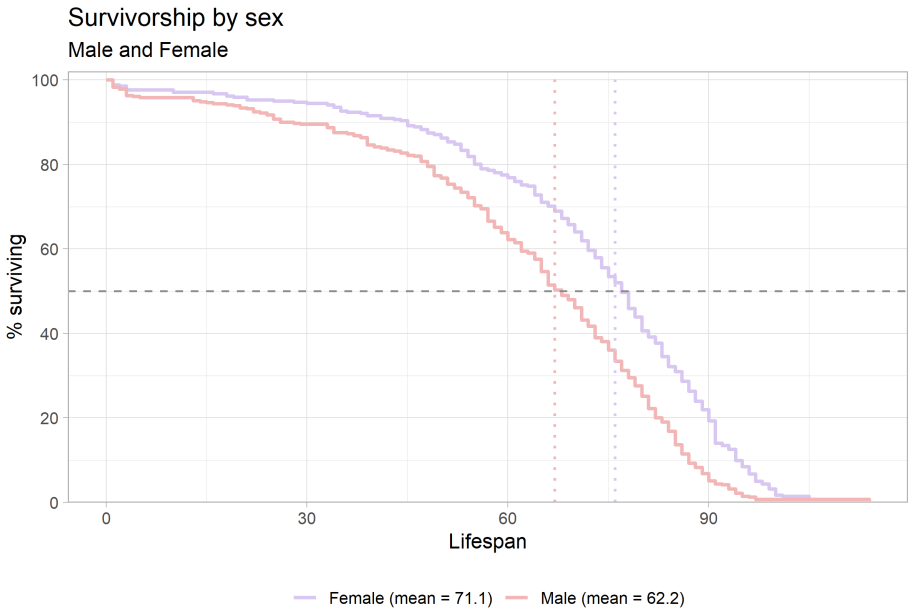


Figure 2: Survivorship curves by sex. The y-axis shows the percentage of the cemetery sample surviving to each lifespan. Dotted vertical lines mark the median lifespan for each sex, where the curve reaches 50%.

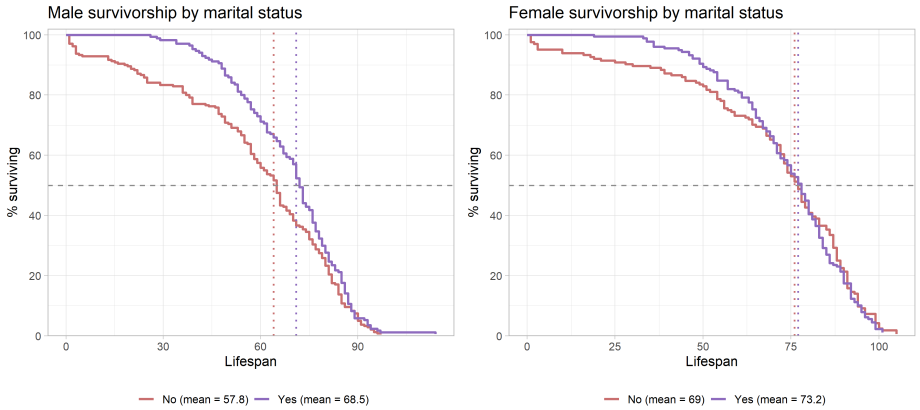


Figure 3: Survivorship by marital status, with separate male and female panels. Dotted vertical lines mark the median lifespan for each marital-status group, where the curve reaches 50%.

water and sanitation contributed substantially to mortality declines in the United States during the twentieth century, especially for infants and children (Cutler & Miller, 2005). The birth-period pattern in this sample [describe results] may be consistent with changing survival conditions, but birth year alone cannot identify which changes mattered. Vaccination, nutrition, medical care, living conditions, and the incomplete representation of people in cemetery records may also contribute to differences across cohorts.

Limitations and next steps: [Discuss sample size, missing or inferred data, cemetery representation, family clustering, survivorship bias, and what additional evidence would strengthen the analysis.]

References

Austad SN, Fischer KE. 2016. Sex Differences in Lifespan. *Cell Metabolism* **23**: 1022–1033.

Cutler D, Miller G. 2005. The role of public health improvements in health advances: The twentieth-century United States. *Demography* **42**: 1–22.

Holt-Lunstad J, Smith TB, Layton JB. 2010. Social Relationships and Mortality Risk: A Meta-analytic Review (C Brayne, Ed.). *PLoS Medicine* **7**: e1000316.

Wang Y, Jiao Y, Nie J, O'Neil A, Huang W, Zhang L, Han J, Liu H, Zhu Y, Yu C, *et al.* 2020. Sex differences in the association between marital status and the risk of cardiovascular, cancer, and all-cause mortality: A systematic

review and meta-analysis of 7,881,040 individuals. *Global Health Research and Policy* **5**: 4.